



Corrigendum to “A practical guide to coding line-by-line trace gas absorption in Earth’s atmosphere” [Journal of Quantitative Spectroscopy & Radiative Transfer 337 (2025) 109345]

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All charts in Fig. 6(c-f) of our paper [1] are missing the parts-per-million scaling factor, $\text{ppm} = 10^{-6}$, on the X-axis for particle number concentration N (molec/cm^3). For example, for H_2O - Fig. 6(c), top charts - the correct X-axis range should be $10^5 \dots 10^{19}$ instead of the shown $10^{11} \dots 10^{25}$ (left chart), and $10^{12} \dots 10^{18}$ instead of the shown $10^{18} \dots 10^{24}$ (right chart).

In addition, the charts in the top row of Fig. 6(d) in [1], which were supposed to show number concentration profiles for ozone (O_3) from MODTRAN [2], mistakenly show those for CO_2 (as in our Fig. 6(c), bottom row). This Corrigendum presents the corrected top charts for Fig. 6(d); X-axis values have also been corrected. The reader may refer to Fig. 2.2 on p.13 in [3] to verify the corrected shape and concentration (X-values) of O_3 in the US 1976 Standard Atmosphere: the red line in our plots corresponds to the grey line in theirs.

These errors have crept only into our Excel file with MODTRAN profiles – a digitized version of that from the PDF [2]. Our code ASPECT

does not read the Excel file; vice versa, the profiles and the ppm scaling factor are hard-coded. The numerical results generated by ASPECT and reported in our paper [1] were not affected by these errors.

The authors would like to apologise for any inconvenience caused.

References

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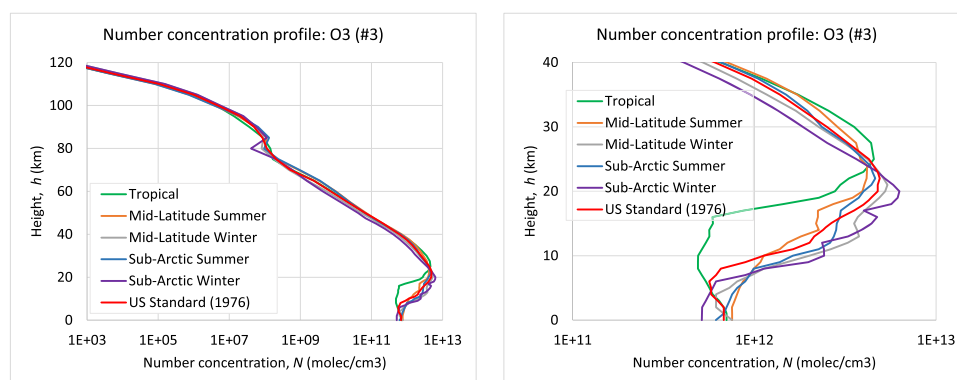


Fig. 6. (d) top charts (corrected): Number concentration of O_3 for 6 MODTRAN atmospheric models. Left: full height range. Right: limited to 40 km.

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